Arash Jalil Khabbazi

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EDUCATION

Purdue University — *PhD, Mechanical Engineering* | GPA 3.9/4.0 West Lafayette, IN | 08/2023 – 08/2028
Concentration in Computational Science and Engineering

- *Research interests:* predictive control, reinforcement learning (RL), smart buildings, energy systems
- *Coursework:* Reinforcement Learning & Control, Applied Optimal Control & Estimation, Convex Optimization, Artificial Intelligence, Statistical Methods, Industrial Internet of Things, Distributed Energy Resources, Analysis of Thermal Systems

University of British Columbia (UBC) — *MS, Mechanical Engineering* | GPA 4.0/4.0 BC, Canada | 05/2021 – 08/2023

University of Tabriz — *BS, Mechanical Engineering* | GPA 4.0/4.0 | Ranked 1 of 155+ EA, Iran | 08/2016 – 08/2020

EXPERIENCE

Purdue University — *Graduate Research Assistant, Kircher Research Group* West Lafayette, IN | 08/2023 – Present

- Designed and ran a three-day demand-flexibility test in a 13-zone campus building, cutting afternoon cooling load **~15%**.
- Derived a multi-zone theorem showing small HVAC demos **overstate savings by up to 100%** (ACC'26, ASHRAE'27).
- Led a 4-institution review of **all known** real-world demonstrations of MPC and RL for HVAC (Applied Energy'25).
- Built load profiles from 25 homes for current-limiting control to **avoid $2k–$10k panel upgrades** (ACEEE'26).
- Developed real-time energy dashboards (Modbus, InfluxDB, Grafana) and a CO_2 occupancy detector with **98% accuracy**.
- Collaborated on a two-parameter, tuning-free HVAC controller with **up to 87% of optimal savings** in simulation (CDC'26).
- Contributed to data-driven predictive control with the **lowest prediction error and failure rate** of all predictors (arXiv'26).

Tesla — *Applied ML Intern* Sparks, NV | 05/2025 – 08/2025

- Developed MySQL data pipelines and engineered high-signal features for defect detection models.
- Trained machine learning (ML) models and ensembles (XGBoost, LightGBM, random forests, SVR, neural nets) to **R^2 0.9**.
- Delivered ML-driven quality gains projected to cut testing costs by over **$1M/year**; offered a **three-month extension**.

Google — *Open-Source Contributor, Smart Buildings Simulator (SBSim)* Remote | 12/2025 – Present

- Co-leading a benchmark of SBSim against EnergyPlus on **RL training speed** with **Google** and **Berkeley Lab**.
- Proposed the **predictive control** workstream, presented **two talks**, contributed code, and **reviewed pull requests**.

Purdue Energy & Utilities — *Energy Systems Researcher* West Lafayette, IN | 02/2026 – Present

- Identified a grey-box thermal model of a 13-zone campus building from BAS data for predictive pre-cooling control.
- Delivered a nine-year energy-use assessment of 21 renovations to **Purdue leadership** for the Giant Leaps Master Plan.
- Quantified a statistically significant **8%** lower chiller approach from an air/dirt-separator retrofit (CampusEnergy'27).

UBC — *Graduate Research Assistant, TMMF Lab & FortisBC Renewable Gas* BC, Canada | 05/2021 – 08/2023

- Ran CFD simulations of hydrogen blending into natural gas pipelines; built ideal- and real-gas mixture models in C.
- Delivered **1** journal paper, **1** conference paper, **2** abstracts, and **2** awards on a project that seeded the UBC H_2Lab.

SELECTED PUBLICATIONS

Full list on Google Scholar

5. W.G. Dierking, **A.J. Khabbazi**, L.D. Reyes Premer, and K.J. Kircher. Scalable supervisory HVAC control for linear objectives. *65th IEEE Conference on Decision and Control (CDC)*, 2026.
4. L.D. Reyes Premer, **A.J. Khabbazi**, and K.J. Kircher. Indirect data-driven predictive control and the state-space predictor. arXiv:2602.10936, 2026.
3. **A.J. Khabbazi** and K.J. Kircher. Small HVAC control demonstrations in larger buildings often overestimate savings. *American Control Conference (ACC)*, 2026.
2. **A.J. Khabbazi**, E.N. Pergantis, L.D. Reyes Premer, P. Papageorgiou, A.H. Lee, J.E. Braun, G.P. Henze, and K.J. Kircher. Lessons learned from field demonstrations of model predictive control and reinforcement learning for residential and commercial HVAC: A review. *Applied Energy*, 2025.
1. L.D. Reyes Premer, E.N. Pergantis, **A.J. Khabbazi**, F. Wu, and K.J. Kircher. Evaluating power control strategies for avoiding residential panel upgrades in cold climates. *ACEEE Summer Study on Energy Efficiency in Buildings*, 2026.

SKILLS

AI Methods: Reinforcement Learning (PPO, SAC), Supervised and Unsupervised Learning, Large Language Models (LLMs)
Control & Optimization: Model Predictive Control, Data-Driven Predictive Control, Convex Optimization, System Identification
Frameworks: CVXPY, TensorFlow, TF-Agents, scikit-learn, XGBoost, Stable-Baselines3, PyTorch, Gurobi
Simulation: Google SBSim, Gymnasium, EnergyPlus, BOPTEST, OCHRE, TRNSYS, EES, Ansys Fluent
Programming & Tools: Python (NumPy, SciPy, Pandas), MATLAB, C, SQL, MySQL, InfluxDB, Grafana, Git, Linux, Docker

SELECT HONORS & AWARDS

ASHRAE Grant-in-Aid Award 2025 | Reidenbach Scholarship, Purdue 2026 | Rakosnik Scholarship, Purdue 2026 | Best PhD Forum Presentation Runner-up, ACM BuildSys 2025 | ACM SIGEnergy Travel Grant 2025 | Best Paper, CSME 2023 | Best Presentation, CSME 2022